International Journal of Research in Industrial Engineering

www.riejournal.com

Int. J. Res. Ind. Eng. Vol. 14, No. 2 (2025) 355–384.

Paper Type: Research Paper

Clustering with Machine Learning and Using NDEA in Development Planning: A Case Study in the Petrochemical Two-Stage SSC

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Citation:

Received: 04 May 2024 *Revised: 03 August 2024* *Accepted: 17 October 2024*	Sayardoost Tabrizi, S., Yakideh, K., Moradi, M., & Ebrahimpour, M. (2025). Clustering with machine learning and using NDEA in development planning: A case study in the petrochemical two-stage SSC. *International journal of research in industrial engineering, 14*(2), 355-384.

Abstract

This research utilizes Machine Learning (ML) to improve the evaluation of Decision-Making Units (DMUs) in a two-stage sustainable supply chain in the petrochemical industry in Iran. Efficiency calculations were conducted for 28 units over 90 time periods. The inputs and outputs of the supply chain were selected according to sustainability criteria, facilitating accurate estimations for production planning and unit development through a hybrid proposed method utilizing both ML and Network Data Envelopment Analysis (NDEA). The goal is to use ML clustering methods alongside network NDEA models to determine the most effective clustering algorithms for categorizing homogeneous units. The primary goal of our research is to leverage ML techniques to enhance the accuracy of decision-making processes, specifically in the clustering of similar units, to assess efficiency. The main goal is to create strategies for improving the performance of inefficient units by comparing them to the most efficient units in each cluster. By implementing the Deep Embedded Clustering (DEC) algorithm, we have discovered substantial improvements in efficiency assessment and development planning. The contrast between clustering outcomes and the traditional NDEA model highlights the importance of clustering in assessing the proximity to the efficient frontier and enabling focused development strategies. This study emphasizes the significance of employing ML for clustering to improve efficiency evaluations and strategic planning for sustainable development in industrial facilities. The results showed that the use of clustering for assessing the relative efficiency of units compared to the non-clustering method with DEA could averagely reduce 18% of the distance of inefficient units from the cluster's efficiency frontier, representing a more attainable ideal goal for inefficient units.

Keywords: Clustering, Machine learning, Sustainable supply chain management, Data envelopment analysis.

1 | Introduction

Data Envelopment Analysis (DEA) is a well-known way to check how well things work in managing and planning supply chains, used a lot for projects to keep things lasting. People often use DEA to measure how well supply chains do, like in the car-making business [1]. It's also used to see if outsourcing work in supply

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https://doi.org/10.22105/riej.2024.455576.1443

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chains is good, looking at how companies perform and the limits in control and seeing what's happening [2]. DEA helps with finding problems and solutions, and understanding green supply chain work [3]. In keeping supply chains lasting, DEA is very important. Reports have used DEA to look at how lasting supply chains are, using moving network ideas [4]. DEA ways have been made to guess which group suppliers in lasting supply chains belong to, helping in picking which suppliers to use [5]. Also, new DEA ways have been put out to judge how lasting supply chains are, focusing on full and part work with wanted and unwanted results [6].

Assessing sustainability in Supply Chain Management (SCM) within industries is a critical area of knowledge that has seen significant advancements through the application of DEA. DEA has emerged as a powerful tool for evaluating different components of SCM, particularly in terms of sustainability. By introducing non-radial DEA models, researchers have been able to assess the environmental efficiency of suppliers, contributing to the selection and evaluation process within SCM [7]. The concept of sustainability in SCM involves meeting a broader set of criteria, often referred to as the triple bottom line approach, which encompasses economic, environmental, and social aspects [8]. Studies have integrated DEA with stochastic double frontier analytic models to evaluate the sustainability and resilience of supply chains, considering factors like netzero and multidimensional features [9]. This integration allows for a comprehensive assessment of sustainability in global supply chains.

DEA has also been utilized in the development of novel models, such as the inverse network dynamic DEA model, to evaluate the sustainability of supply chains by determining the point at which the Local Best Decision (LBD) occurs [10]. These models provide insights into the efficiency and sustainability of supply chain Decision-Making Units (DMUs) through dynamic network DEA approaches. In summary, the application of DEA in assessing sustainability in SCM has led to significant advancements in evaluating environmental efficiency, resilience, and overall sustainability performance within supply chains. By integrating DEA with various modeling approaches and frameworks, researchers have been able to provide valuable insights into enhancing sustainability practices in industries through effective SCM.

Using closed-loop systems in the state-of-the-art model based on the Circular Economy (CE) concept and the efficiency assessment of DMUs using DEA is crucial for sustainable resource management. The CE emphasizes closed loops for materials, focusing on recovering and recycling to keep resources circulating within the economy [11]. This concept involves minimizing resource input and waste by closing material and energy loops. To implement CE strategies effectively, closed-loop thinking requires innovation and adaptation [12].

Undesirable outputs are a crucial component in the evaluation of efficiency using Network Data Envelopment Analysis (NDEA) models. Undesirable outputs are defined as outputs that have negative impacts on the system or environment. In the context of NDEA models, these outputs represent the negative consequences or by-products of production processes that must be minimized or eliminated to enhance overall efficiency. NDEA models are specifically designed to incorporate both desirable and undesirable outputs when assessing DMUs. This approach allows for a comprehensive evaluation of efficiency by considering the impact of both types of outputs on the performance of DMUs [13]. By integrating undesirable outputs into the analysis, NDEA models offer a more accurate depiction of efficiency, capturing the trade-offs between desirable and undesirable aspects of production processes [14].

The presence of undesirable outputs in NDEA models requires a balanced approach that focuses not only on maximizing desirable outputs but also on minimizing undesirable outputs to improve overall efficiency. By including undesirable outputs in the evaluation framework, NDEA models provide a holistic view of efficiency, considering the full spectrum of outputs generated by DMUs [15]. In conclusion, the incorporation of undesirable outputs in NDEA models is essential for a comprehensive assessment of efficiency and the promotion of sustainable practices. By accounting for both desirable and undesirable outputs, NDEA models offer valuable insights into the efficiency of DMUs, empowering organizations to make informed decisions to optimize their processes while mitigating negative impacts on the environment and society.

In countries with natural resources of oil and gas, the process of extracting crude oil and transporting it to the refineries is typically managed through a multi-level supply chain. In this study, a two-level supply chain model is proposed, consisting of the oil extraction stage (Stage 1) and the refining stage (Stage 2), and assessed using the efficiency evaluation model based on NDEA. Each petrochemical unit in the supply chain is considered a DMU in the network model, and the inputs and outputs at the beginning and end of the network, as well as the exchanges between the two levels of the supply chain, are all taken into consideration when determining the efficiency [16].

Improving operational efficiency and fairly evaluating decision units present significant obstacles in SCM . To overcome these hurdles, it is crucial to categorize similar and comparable units into clusters and then perform DEA calculations for each cluster individually. This approach allows the most efficient unit within each cluster to function as an exemplary model for the inefficient units. On the other hand, the most efficient unit of each cluster will be able to move to the higher cluster and be ready to compete to improve its efficiency and sustainability with its decision units in future periods. The use of clustering methods was placed on the agenda [17]. In similar research using an artificial neural network, decision units were placed in homogeneous clusters and efficiency calculation was evaluated using efficient and inefficient boundary methods [18]. Utilizing the applications of artificial Neural Networks (NN) is not merely subject to clustering, as in the paper [19]; it has been used alongside dynamic and network envelopment analysis to predict sustainability in the future. Using artificial intelligence to predict efficiency to provide a solution for over-inefficiency improvement is one of the most state-of-the-art methods in the field of sustainable supply chains. Our novelty is using Machine Learning (ML) for clustering two-stage SSC of the petrochemical industry in Iran for the first time and proposing a new method for assessing efficiency with NDEA. This section can also be named as one of the knowledge gaps identified in this research. In summary, the contributions of the study are as follows:

I. We proposed a hybrid approach with NDEA and ML for clustering the supply chain of petrochemical industries.

II. We compared the efficiency of the two-stage sustainable supply chain in the presence of undesirable output and without the presence of them.

III. We provide a new comparable clustering method to find the best clustering algorithms for our dataset.

IV. For the first time, we investigated the impact of clustering with ML algorithms on calculating the efficiency of SSC to provide the improvement solution and development plan for inefficient DMUs.

The rest of this paper is organized as follows. A literature review is presented in Section 2. In Section 3, the proposed method is given. A case study is presented in Section 4. Finally, managerial implications and concluding remarks are discussed in Sections 5 and 6.

2 | Literature Review

Articles related to this research are divided into three main sections. According to the title of the paper, the main priority was to calculate the efficiency of grouping DMUs into homogeneous clusters. Therefore, the main center focused on the study of algorithms in the context of ML After completing the clustering using network data enveloping analysis, inputs and outputs were examined sustainable approach in the two-stage supply chain of the petrochemical industry in Iran. In this way, for main parts of the research background are: 1) the application of ML in SCM, 2) the sustainability of supply chains, 3) network envelopment analysis, and 4) the knowledge gap.

2.1 | Application of Machine Learning in Supply Chain Management

AI offers many benefits in operations and SCM, such as improved productivity, reduced errors, improved processes, and increased productivity. AI has a significant impact on business processes through real-time analysis, predictive maintenance and quality control. Implementation challenges include identifying needed capabilities, addressing cost issues, and navigating potential barriers to successful AI integration. In spite of

these challenges, the potential that AI implementation has to deliver productivity gains, enhance competitive advantage, and increase operational efficiency remains promising in operations and SCM [20].

The fourth technological revolution is ML, which has increased decision-making accuracy and replaced the human brain in important intuitive decisions where speed and accuracy are important factors. In this case, data collection enables the machine to learn like experience. With the ability to learn from data and high processing speeds, the machine will be able to recognize complex patterns and relationships between different variables to make the best decision. In the field of SCM, the use of ML in multi-criteria decision-making under conditions of uncertainty, based on the ability to recognize non-linear relationships, will lead to accurate predictions and the prevention of the whiplash effect, ultimately increasing the overall efficiency of the supply chain. SCM uses ML to enhance efficiency, make better choices, and streamline operations. ML algorithms like Decision Trees (DT), NN, and Support Vector Machines (SVM) are frequently employed in SCM activities like demand forecasting and sales estimation, aiding in more effective and efficient supply chain operations. However, there are challenges in implementing ML in SCM, including the scarcity of highly skilled authors, disparities in the distribution of ML algorithms across SCM functions, cost considerations, and data suitability issues. The ongoing research in ML applications within SCM aims to provide valuable insights for future SCM practices and research directions despite these challenges [20]. Reviewing previous research showed that although the issue of sustainable supply chain for the petrochemical industry has been investigated with different methods and approaches, the application of ML in the field of decision unit clustering has not been considered by researchers. Then, by studying the quality of clustering of several ML sub-algorithms, the best method of clustering was determined using ML . In this section, a report on the types of ML algorithms that were used in the field of supply chain between 1998 and 2018 from 123 articles that are visible in *Fig. 1*.

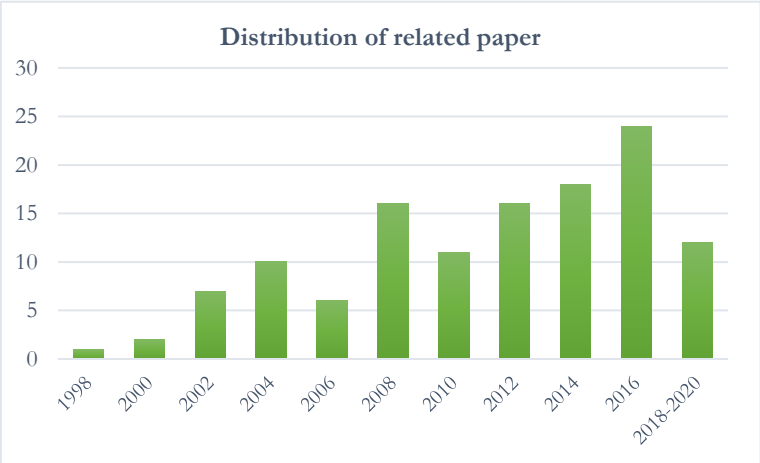


Fig. 1. Distribution of published articles in the field of ML related to supply chain.

Between 2008 and 2009, the remarkable growth in the use of ML in supply chain research coincided with the financial crisis, when its learning and predictive capabilities were utilized to manage supply and demand among the chain participants. The widespread use of soft computing in interdisciplinary research coincided with the second peak in 2016-2017. Over the past two decades, *Fig. 2* reveals the variety of ML algorithms employed in the field of supply chain administration. The tendency of researchers to employ neural networks in prediction has always been considered, but now, thanks to the creation of data warehouses in organizations, the phenomenon of large data that is caused by the existence of multiple databases in industries and organizations is more than apparent [21]. Data mining and ML can be very effective in clustering and classification. *Fig. 2* shows that more than 75% of published papers used only neural networks and SVM algorithms for various purposes, often predicted. The other category revealed that 81 percent of the 20-year-old published articles dealt with supervised learning, 12% with unsupervised learning, and 7% with reinforcement learning. From reviewing these statistics, it is concluded that it is very limited to address the

algorithms of ML sub-discipline, especially unsupervised learning, in which clustering is one of the most important in the field of SCM, and a very suitable field is introduced for researchers. In *Fig. 2*, the classification of ML algorithms was shown with the frequency of usage in articles.

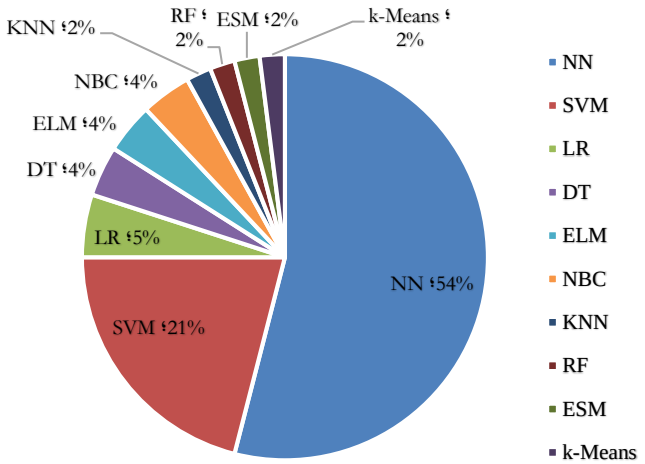


Fig. 2. Classification of ML algorithms based on the amount of use in articles.

2.2 | Sustainability of Supply Chains

The supply chain has a complex multi-layered network. On the other hand, supply chains consist of departments or organizations that produce and distribute goods to end customers. Adding value to all stakeholders involved in the supply chain is the ultimate goal of every company. In evaluating the efficiency of a DMU, it is important to address various steps. Sustainability of the supply chain includes economic benefits, social responsibility, and environmental protection [22]. Sustainability implies a fine balance between economic benefits, environmental protection, and institutional social responsibility. In order to be sustainable, companies strive to produce environmentally and socially friendly products. Furthermore, all sustainability standards should be considered sustainable. Sustainability plays an important role in achieving sustainable supply chains [23].

The SSCM refers to the consideration of environmental, social, and economic criteria when evaluating supply chains [24]. In recent years, sustainability has become an essential part of SCM. Focusing on greener products and services in the SCM is a response to the pressures from authorities, environmentalists, customers, and Non-Governmental Organizations (NGOs). Therefore, communities, businesses, and NGOs are looking for tools to measure the sustainability progress of organizations [25]. In [25], authors conducted a literature review on SSCM and discussed possible implications for future research works. Furthermore, sustainability is a key issue for researchers because of the decline in natural resources and concerns regarding capital disparity and societal obligations [26].

The [27] discussed that concerns about sustainability have increased the accountability and responsibility of supply chains. They proved a significant relationship between the various aspects of sustainability and firm performance. Recently, Alrasheedi et al. [28] suggested a hybrid fuzzy decision-making framework to assess the sustainability of suppliers in manufacturing firms. Furthermore, in an attempt to justify the implementation of Blockchain, Yadav and Singh [29] developed an integrated fuzzy-ANP framework for assessing sustainable supply chains. The [25] conducted a literature review on SSCM and discussed possible implications for future research works. In [27], authors used the SSCM concept to evaluate sustainable suppliers via the congestion approach of the economies of scale theory of DEA. Other related papers improved the SSCM using a hierarchical grey-Decision Making Trial and Evaluation Laboratory

(DEMATEL) approach. DEMATEL is an applicable tool used for assessing interdependent interactions among factors to identify cause-effect chain components of a complicated network [30].

Organizations are responsible for their environmental and social performance. Sustainable development emphasizes on environment, economy, and society. Evaluating Sustainable Supply Chain Management (SSCM) performance is a key challenge. SSCM has turned out to be a strategic process that enables firms to create a competitive advantage. Customers' demands and products' complexity have increased competition among organizations. SSCM provides an opportunity for organizations to distinguish itself from its competitors. Thus, it provides a competitive advantage in the market. Numerous organizations have already commenced to develop a certain level of commitment toward sustainability practices to make their supply chains sustainable. SSCM evaluation, hence, is a substantial duty for any organization. Among evaluation methods, DEA appears to be a suitable technique to assess SSCM.

2.3 | Network Data Envelopment Analysis

Today, due to the global shape of markets and increasing competition between supply chains, paying attention to sustainability criteria is an inevitable task for all businesses. A two-stage DEA network model was proposed in [31] for assessing Sustainability Efficiency (SE) and Profitability Efficiency (PE) of Mining Multinational Enterprises (MNEs). Higher sustainability is associated with increased returns for companies, showing that improvements in SE can lead to better PE. A framework was introduced in this method besides the multistage model with NDEA. Our purpose is to use DEA in the petrochemical industry to measure efficiency with DEA to evaluate the performance of multistage units in the petrochemical industry supply chain in Iran. This model allows for a more comprehensive analysis of efficiency within the supply chain of petrochemical companies. By considering the interconnectedness of multistage units, the NDEA model offers a more accurate evaluation of overall efficiency and the efficiency of individual components.

The primary goal of [32] is to provide an inverse DEA model for a two-stage network-structured production system in the presence of undesired factors with weak disposability, given the significance of network-structured production systems in real life. The suggested model's main goal is to estimate the amount that one or more process indicators would change if another stage's indicators were changed while maintaining the same degree of efficiency. The ability of the suggested procedure to simultaneously lower input levels and raise output levels is by far its greatest benefit.

However, one of the most growing criticisms in these sorts of models is that the DMUs are assessed in a specific period. Given fluctuations in the performance of DMUs during several periods, considering just a specific period is insufficient in comprehensive efficiency evaluation. In dynamic DEA presented by Tone and Tsutsui [16], the DMUs' efficiencies are assessed during several periods. However, dynamic DEA does not deal with DMUs' internal structures. NDEA deals with DMUs with internal. Tone and Tsutsui [16] proposed a network-dynamic DEA model in which the DMUs were assessed in multiple periods with internal interactions. Their model is based on the Slack-Based Measure (SBM) model. In other words, NDEA is one of the advanced DEA models for dealing with the network structure of DMUs. A wide range of NDEA models has been used for assessing and ranking DMUs. The NDEA can rank DMUs with network structures. In [33], authors proposed a multiplicative relational NDEA model in the presence of window analysis to capture the efficiency trends over time. However, they did not present a robust optimization model. The classical DEA models assume deterministic inputs and outputs. In other words, the classical methods for considering the uncertainty of parameters are divided into two categories: sensitivity analysis and random optimization. In sensitivity analysis, the uncertainty is initially ignored. Then, after solving the problem, the effect of uncertain data is checked [34].

The other study investigates the fiscal performance and efficiency of Indian states from 2009 to 2015 using a NDEA approach, focusing on tax mobilization and development spending. The findings reveal a positive correlation between efficiency scores and GDP growth as well as gross capital formation while highlighting a negative association between efficiency and outstanding liabilities. This emphasizes the necessity for balanced

governmental spending strategies to prevent future debt crises in India. This two-stage methodology allows for an in depth analysis of fiscal efficiency in tax and spending processes at a granular level. A significant negative correlation exists between outstanding liabilities and efficiency scores, highlighting fiscal sustainability concerns. Finally, the study underscores the importance of effective financial planning, combining tax financing and quality expenditure management for sustainable growth [35].

The potential DEA has for being used as a useful instrument in tackling sustainability concerns in SCM. The results highlight the growing trends in the field and emphasize the role of the conceptual approaches that address the tensions between business performance and excellence in sustainable practices. The Key points of this paper are:

I. About 352 articles focusing on the DEA in SCM, and the number of citations reached 10,445, illustrating the increasing importance of the field. The majority of the analyzed papers were published in Business Economics and Engineering, which proves the best behavior of the literature in the further investigation of the context.

II. The majority of citations for DEA in SCM relate to incorporating neural networks and multi-criteria decision-making for green supplier selection.

III. As one would expect, there has been a surge of demand for effective supply chain sustainability in the most recent years, where DEA has been recognized as one of the key tools in measuring sustainability [36].

In past research, organizational performance assessment primarily focused on economic and financial indicators; however, a shift has emerged toward recognizing essential cognitive and social parameters that meet scientific standards, driven by growing human needs. The Balanced Scorecard (BSC) framework seeks to balance past-oriented financial metrics with future-oriented dimensions such as customer processes, learning and growth, and the development of human and social capital. DEA serves as a non-parametric method for measuring efficiency across similar units with different inputs and outputs, with a two-level DEA approach applicable for classifying inputs and using common weights in cases of multiple similar variables. This paper aims to utilize the BSC for designing performance assessment indicators and the two-level DEA for effective measurement, acknowledging the expansion and diversification of inputs in today's complex social systems [37]. Measuring the performance of production systems is a vital management function, effectively addressed by the BSC framework, which offers a structured visualization of key performance indicators across financial, customer, internal process, and learning and growth dimensions.

In contrast, conventional DEA methods assess systems as a whole, failing to evaluate individual processes. This paper introduces an integrated BSC-DEA methodology that facilitates multi-criteria performance evaluation across various project stages. The combined BSC-DEA model employs cards to categorize input and output metrics linked to BSC perspectives, allowing for efficiency decomposition to pinpoint specific processes that contribute to systemic inefficiencies and guide future improvements. The authors illustrate this integrated BSC-DEA framework through a case study of six banking branches, demonstrating its ability to provide a more comprehensive assessment of organizational performance compared to traditional methods [38].

The other model focuses on a two-warehouse system for perishable items, employing minimal total cost strategies during COVID-19 lockdowns. This study presents an inventory model designed to minimize costs in a two-warehouse system while accounting for deterioration rates influenced by time and disruptions in demand due to the COVID-19 lockdown. Two distinct scenarios are analyzed: one with stock depletion after lockdown and another with depletion during lockdown. The study incorporates time-dependent demand and deterioration rates represented using weibull distributions. Utilizing a Last In, First Out (LIFO) policy helps reduce overall holding costs in the higher-cost rented warehouse. The research includes sensitivity analysis, indicating that various factors, such as demand fluctuations and warehouse conditions, significantly impact costs. Numerical examples demonstrate practical applications of the model, confirming the higher total costs

associated with longer lockdown periods. The findings advocate for enhanced capacity and strategic use of owned warehouses to minimize holding and deterioration costs effectively [39].

2.4 | Research Gap

To the best of our knowledge, there is no article to evaluate sustainable supply chains using ML for Clustering and two-stage NDEA in the presence of undesirable outputs especially in petrochemical industries in Iran. This paper presents an approach to measure sustainability using both desirable and undesirable outputs. One of the knowledge gap insights in this article is the importance of grouping similar industrial units in on cluster and making plans to develop inefficient units to reach the most efficient unit of each cluster.

One of the key challenges in comparing relative efficiency using traditional NDEA lies in benchmarking the performance of all units against the most efficient units on the efficiency frontier. This approach is logically flawed, as it fails to account for the varying capacities and specific capabilities of petrochemical units in Iran. A more effective strategy involves classifying homogeneous units into specific clusters, where the units share similar input and output volumes. Moreover, the application of ML in the clustering of supply chains remains underexplored. This article addresses this gap by proposing a method for selecting the most appropriate algorithm based on the dataset under investigation and offers a framework for comparing algorithms to identify the optimal one based on clustering quality. Consequently, this research not only demonstrates the process of selecting the best algorithm but also highlights the impact of using ML in sustainable supply chain clustering. Additionally, this approach offers valuable insights for planning the developmental strategies of industrial units, and our proposed hybrid method using ML and NDEA can be implemented for various industries; this article has examined a case study for the proposed method.

3 | Proposed Method

In summary, to introduce our proposed approach, the following steps are taken: In *Step 1*, DMUs are clustered using three distinct algorithms within the realm of ML. In *Step 2*, The clustering outputs of three different algorithms were compared using clustering quality evaluation indicators, resulting in the identification of the best clustering algorithm for the dataset of our case study. In *Step 3*, evaluate the efficiency of the DMUs in each cluster separately and identify the most efficient unit in the cluster. In *Step 4,* the results from the previous step were compared with the conventional DEA method. In the last step, the results showed that assessing the efficiency among homogeneous DMUs in each cluster can lead inefficient DMUs to plan achievable development plans. These steps are presented below:

Step 1. DMUs are clustered using three distinct ML algorithms: by using Python programming language, three different algorithms will be implemented on the dataset of our case study.

Step 2. Selecting the best clustering method: by comparing the results of three clustering algorithms with each other, in this step, the clustering quality of the three different algorithms will be compared based on mathematical scores with each other.

Step 3. Calculating efficiency in each cluster by NDEA model: evaluating the efficiency of DMUs in each cluster separately and identifying the most efficient DMU in each cluster as the best unit.

Step 4. Comparing the proposed method with the traditional NDEA approach: all results from our proposed method based on two-stage NDEA will compared with traditional NDEA, which is calculating efficiency between all DMUs without clustering.

Step 5. Discussion and development planning: analysis and discussion is the final step. In this step, the efficient unit will be considered as a target for other units in each cluster.

3.1 | Two-Stage NDEA

In the initial DEA, the internal structures of DMUs are ignored. The NDEA assesses the efficiency of the internal structure of DMUs. In other words, NDEA is one of the advanced DEA models for dealing with

the network structure of DMUs. The classical DEA models assume deterministic inputs and outputs. In other words, the classical methods for considering the uncertainty of parameters are divided into two categories: sensitivity analysis and random optimization. In sensitivity analysis, the uncertainty is initially ignored. Then, after solving the problem, the effect of uncertain data is checked. Before proposing the model, the used notations are introduced. The used notations are listed in *Table 1*.

Table 1. The notations.

Notations	Definitions
x_{ij}	The ith input of DMUj; (i = 1, … ,m; j = 1, …,n)
y_{rj}	The rth output of DMUj; (r = 1, … ,s; j = 1, …,n)
z_{dj}	The dth intermediate index of DMUj; (d = 1, … ,p; j = 1, …,n)
v_i	The weight of the x_{ij}
w_d	The weight of the y_{rj}
u_r	The weight of the z_{dj}
E_{1o}	The efficiency score of the first stage
E_{2o}	The efficiency score of the second stage
E_o	The overall efficiency score
w_1	The semi-positive weight of the first goal, which is determined by the decision-maker.
w_2	The semi-positive weight of the second goal, which is determined by the decision-maker.
x_{ij1}	The ith input of the first stage of DMUj; (i = 1, … , m; j = 1, …,n)
x_{kj2}	The kth input of the second stage of DMUj; (k = 1, … , K; j = 1, …,n)
y_{rj}^{2g}	The rth desirable output of the second stage of DMUj; (r = 1, … ,s; j = 1, …,n)
y_{hj}^{2b}	The hth undesirable output of the second stage of DMUj; (h = 1, … , H; j = 1, …,n)
v_k	The kth weight of the input of the second stage; (k = 1, … , Q)
u_r	The rth weight of the output of the second stage; (r = 1, … , S)
μ_h	The hth weight of the undesirable output of the second stage; (h = 1, … ,H)

However, it is not a good way to generate robust responses to data changes. In robust optimization, which is a kind of stochastic optimization approach, we are looking for near-optimal solutions. Here, we introduce a two-stage NDEA model to assess the efficiency of DMUs [34]. As shown in *Fig. 3*, the first stage of DMUj uses x_{ij} (i = 1, …, m) to produce z_{dj} (d = 1,…, p). The outputs of the first stage (z_{dj}) are the intermediate measures that are used as inputs in the second stage to produce final outputs y_{rj} (r = 1, …, s). The two-stage structure is illustrated in *Fig. 3*. The basic input-oriented Constant Returns to Scale (CRS) model, which independently calculates the efficiency of stages 1 and 2, is as follows:

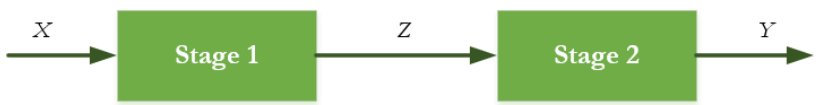


Fig. 3. The two-stage network.

$$
\begin{aligned}
&E_o^1 = \max \sum_{d=1}^{p} w_d z_{do}, \\
&\text{s.t.} \\
&\sum_{i=1}^{m} v_i x_{io} = 1, \\
&\sum_{d=1}^{p} w_d z_{dj} - \sum_{i=1}^{m} v_i x_{ij} \le 0, \quad j = 1,\dots,n, \\
&v_i \ge 0, w_d \ge 0, (i = 1,\dots,m; t = 1,\dots,p).
\end{aligned} \tag{1}
$$

$$
\begin{aligned}
&E_o^2 = \max \sum_{r=1}^{s} u_r y_{ro}, \\
&\text{s.t.} \\
&\sum_{d=1}^{p} w_d z_{do} = 1, \\
&\sum_{r=1}^{s} u_r y_{ro} - \sum_{d=1}^{p} w_d z_{do} \le 0, \quad j = 1,\dots n, \\
&u_r \ge 0, w_d \ge 0, (r = 1,\dots,s; d = 1,\dots,p).
\end{aligned} \tag{2}
$$

According to the *Fig. 3* and the *Eqs. (1)* and *(2)*, models can be considered as follows:

$$
\begin{aligned}
&\bar{E}_o^1 = \max k_1, \\
&\text{s.t.} \\
&-\sum_{d=1}^{p} w_d z_{do} \le -k_1, \\
&\sum_{i=1}^{m} v_i x_{io} \le 1, \\
&\sum_{d=1}^{p} w_d z_{dj} - \sum_{i=1}^{m} v_i x_{ij} \le 0, \ j = 1,\dots,n, \\
&v_i, w_d \ge 0, \ \text{ for all i, for all d.}
\end{aligned} \tag{3}
$$

$$
\begin{aligned}
&\bar{E}_o^2 = \max k_2, \\
&\text{s.t.} \\
&-\sum_{r=1}^{s} u_r y_{ro} \le -k_2, \\
&\sum_{d=1}^{p} w_d z_{do} \le 1, \\
&\sum_{r=1}^{s} u_r y_{ro} - \sum_{d=1}^{p} w_d z_{do} \le 0, \quad \text{for all j,} \\
&u_r \ge 0, w_d \ge 0 \quad \text{(for all r, for all d).}
\end{aligned} \tag{4}
$$

Lemma 1. The optimal values of *Models (1)* and *(3)* are equal.

Proof: Note that the feasibility set of *Model (1)* is a subset of the feasibility set of *Model (3)*. Then, $E^1_o \le \bar{E}^1_o$. On the other hand, given that *Model (3)* is maximization, the best condition for the constraints $\sum_{i=1}^{m} ViXio \le 1$ and $-\sum_{t=1}^{p} WdZdo \le -k1$ is the equality sign. Given that the rest of the constraints are the same in both models, $E^1_o \ge \bar{E}^1_o$. As a result, the optimal value of both models is equal, and the lemma is proved.

Lemma 2. The optimal values of *Models (2)* and *(4)* are equal.

Proof: according to *Lemma 1*, the proof of this lemma is obvious.

In DEA, the goal is to get the best efficiency score for DMUs. Here, we follow a similar approach. We wish to maximize the efficiency of each stage to get the best overall efficiency score in a two-stage network. Our proposed model measures the efficiency score of two-stage networks by the following two objective functions:

$$
\begin{aligned}
&\bar{E}_o^1 = \max k_1, \\
&\bar{E}_o^2 = \max k_2, \\
&\text{s.t.} \\
&-\sum_{d=1}^{p} w_d z_{do} \le -k_1, \\
&-\sum_{r=1}^{s} u_r y_{ro} \le -k_2, \\
&\sum_{i=1}^{m} v_i x_{io} \le 1, \\
&\sum_{d=1}^{p} w_d z_{do} \le 1, \\
&\sum_{d=1}^{p} w_d z_{dj} - \sum_{i=1}^{m} v_i x_{ij} \le 0, \quad j = 1,\ldots n, \\
&\sum_{r=1}^{s} u_r y_{ro} - \sum_{d=1}^{p} w_d z_{do} \le 0, \quad j = 1,\ldots n, \\
&u_r \ge 0, v_i \ge 0, w_d \ge 0, \qquad \text{for all r, for all i, for all d.}
\end{aligned} \tag{5}
$$

To solve *Model (6)*, we can use the weighted method. Consequently, we get the following model with only one objective function:

$$
\begin{aligned}
&E_o = \max (w_1 k_1 + w_2 k_2), \\
&\text{s.t.} \\
&-\sum_{d=1}^{p} w_d z_{do} \le -k_1, \\
&-\sum_{r=1}^{s} u_r y_{ro} \le -k_2, \\
&\sum_{i=1}^{m} v_i x_{io} \le 1, \\
&\sum_{d=1}^{p} w_d z_{do} \le 1, \\
&\sum_{d=1}^{p} w_d z_{dj} - \sum_{i=1}^{m} v_i x_{ij} \le 0, \quad j = 1,\ldots,n, \\
&\sum_{r=1}^{s} u_r y_{ro} - \sum_{d=1}^{p} w_d z_{do} \le 0, \quad j = 1,\ldots,n, \\
&u_r, v_i, w_d \ge 0, \qquad \text{for all r, i, d,}
\end{aligned} \tag{6}
$$

where w_1 and w_2 are the weights of the first and second objective functions, respectively, the weights should be semi-positive, and the sum of them should be unity. Decision-makers determine the weights. Assume that

(v^*_i, w^*_d, u^*_r) are the optimal answers for *Model (6)*. The efficiency of the stages and the overall efficiency are calculated as follows:

$$\bar{E}_o^1 = \frac{\sum_{d=1}^{p} w_d^* z_{dj}}{\sum_{i=1}^{m} v_i^* x_{ij}}, \bar{E}_o^2 = \frac{\sum_{r=1}^{s} u_r^* y_{ro}}{\sum_{d=1}^{p} w_d^* z_{dj}}, E_o = w_1 \frac{\sum_{d=1}^{p} w_d^* z_{dj}}{\sum_{i=1}^{m} v_i^* x_{ij}} + w_2 \frac{\sum_{r=1}^{s} u_r^* y_{ro}}{\sum_{d=1}^{p} w_d^* z_{dj}},$$

or

$$E_o = \bar{E}_o^1 * \bar{E}_o^2.$$

As a result of the above, the overall efficiency score is equal to the weighted average of stages. Another evaluation of the overall efficiency score can be calculated by multiplying the amount of efficiency score of stage 1 and stage 2. All results have been compared in Section 4 of the paper.

3.2 | Two-Stage NDEA Model in the Presence of Undesirable Outputs

Assume that there are n DMUs, which consume m inputs to produce s desirable outputs and h undesirable outputs. According to Seiford and Zhou [40], the output-oriented fractional model in the presence of undesirable outputs is as follows:

$$\begin{aligned} & E_o^* = \min \frac{\sum_{i=1}^{m} v_i x_{io} + v_o}{\sum_{r=1}^{s} u_r y_{ro} + \sum_{h=1}^{H} \mu_h \bar{b}_{ho}}, \\ & \text{s.t.} \\ & \frac{\sum_{i=1}^{m} v_i x_{ij} + v_j}{\sum_{r=1}^{s} u_r y_{rj} + \sum_{h=1}^{H} \mu_h \bar{b}_{hj}} \geq 1, \qquad j = 1, \cdots, n, \\ & u_r, v_i, \mu_h \geq 0 \quad \text{and} \quad v_o \text{ free in sign.} \end{aligned} \tag{8}$$

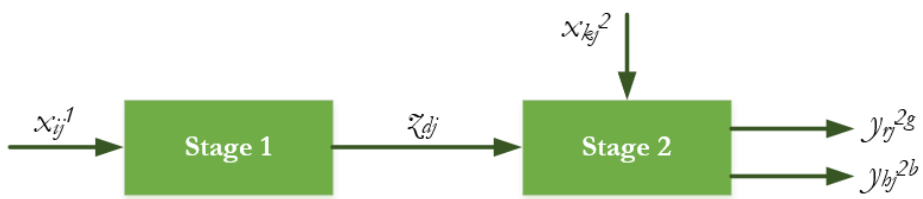


Fig. 4. The two-stage network with both desirable and undesirable outputs.

In this model, the undesirable outputs are transformed using a linear transformation (b_{tj}), where the alpha (α) value is set so high to keep all values in The range of positive numbers. Alpha is a big number, which is called a bias value, and it is a big enough number to ensure that the value of y_{hj}^{2b}(hat) is positive., so the transformed value is positive. Other parameters like v_i and μ_t are weights of input and undesirable output, respectively. *Fig. 4* shows a two-stage network such that each stage has desirable and undesirable outputs. As seen in *Fig. 2*, in the first stage, the input x_{ij}^1 is used to produce the output z_{dj} (intermediate output or the output of stage 1). In the second stage, the input z_{dj} and x_{kj}^2 are used to produce the desirable outputs as y_{rj}^{2g} and undesirable output as y_{hj}^{2b}. Here, z_{dj} is the intermediate measure, b is the undesirable output, and g is the desirable output, which is used in this model that is illustrareted in *Fig. 4*. We change the variables as follows:

$$\bar{y}_{hj}^{2b} = -y_{hj}^{2b} + \alpha, \quad (j = 1, 2, ..., n).$$

Based upon the presented models in the previous section, we show the following model to assess the efficiency of DMU_o in the presence of undesirable output:

$$
\begin{aligned}
& E_o^* = \max \quad (w_1 k_1 + w_2 k_2), \\
& \text{s.t.} \\
& -\sum_{d=1}^{D} w_d z_{do} \le -k_1, \\
& -\sum_{r=1}^{R} u_r y_{ro}^{2g} - \sum_{h=1}^{H} \mu_h \bar{y}_{ho}^{2b} - v_o \le -k_2, \\
& \sum_{i=1}^{m} v_i x_{io}^1 \le 1, \\
& \sum_{d=1}^{D} w_d z_{do} + \sum_{k=1}^{K} v_k x_{ko}^2 \le 1, \\
& \sum_{d=1}^{D} w_d z_{dj} - \sum_{i=1}^{m} v_i x_{ij} \le 0, \quad j = 1, 2, \ldots, n, \\
& (\sum_{r=1}^{s} u_r y_{rj}^{2g} + \sum_{h=1}^{H} \mu_h \bar{y}_{hj}^{2b} + v_o) - (\sum_{d=1}^{D} w_d z_{dj} + \sum_{k=1}^{K} v_k x_{kj}^2) \le 0, \quad j = 1, 2, \ldots, n, \\
& \mu_h, w_d, u_r, v_k, v_i \ge 0 \quad \text{and} \quad v_o \text{ is free in sign.}
\end{aligned}
$$

This model is a multiplicative model in which the weights are unknown. Suppose that all parameters are the optimal solution of *Model (9)*. The efficiency score of each stage and the overall efficiency of DMUo are the same as *Model (7)*. We can change the amount of w_1 and w_2 by the impact of each stage on a specific supply chain. In our case study, the results of overall efficiency have been compared between the average method and the multiplicative method.

4 | Case Study: Petrochemical Sustainable Supply Chain in Iran

The oil and gas industry is one of the most crucial for Iran. It is one of the largest producers of oil and natural gas in the world and has access to vast reserves of both resources. Moreover, the petrochemical industry in the country is also significant. Among the products that the industry manufactures are plastics, fertilizers, and various chemicals; they find application in agriculture, manufacturing, and construction. Several plants are producing these goods in Iran, which belong to both governmental and private companies. For the petrochemical industry in Iran, SCM is an essential component since it integrates sourcing, production, and the movement of raw materials and finished products at various levels. Iran's petrochemical sector produces a wide array of products, and sourcing and transportation to numerous countries is only possible with the coordination brought by acquiring the necessary resources and moving products to the foreign market. Finally, effective SCM is essential for the success of the petrochemical industry in Iran. By optimizing processes, reducing costs, and improving efficiency, companies can enhance their competitiveness and contribute to the growth and development of the industry as a whole.

In a two-stage supply chain model in the petrochemical industry, with the first stage being extraction and the second stage being refining, incorporating sustainability practices is crucial for ensuring environmental responsibility, social impact, and economic viability.

Extraction stage

In the extraction phase, companies can implement sustainable practices such as B. Minimizing environmental impact, reducing water consumption, and handling waste responsibilities. This may include using advanced drilling and production technologies, implementing water recycling systems, and adhering to strict environmental regulations. Companies can work with local communities near mining sites to ensure their

operations are socially responsible and benefit local communities. This may include providing employment opportunities, supporting local infrastructure development, and advising community members on environmental and social issues. In addition to the mentioned cases, controlling the HSE unit to increase the safety of personnel is another necessary measure in the field of social responsibilities in the criteria of sustainability.

Refining stage

In the refining phase, DMUs can focus on improving energy efficiency and reducing greenhouse gas emissions. This may include investing in energy-efficient technologies, optimizing production processes and using renewable energy sources where possible. Refineries can also implement waste management practices to minimize waste generation, recycle materials, and safely dispose of hazardous waste. This can help reduce environmental impact and comply with regulatory requirements. By integrating sustainability principles into both the extraction and refining phases of the petrochemical supply chain, companies can reduce their environmental footprint, increase their social impact and improve their long-term profitability. This approach not only benefits the company but also contributes to the overall sustainability of the petrochemical industry in Iran.

4.1|Two-Stage Sustainable Supply Chain in the Petrochemical Industry

Crude oil is extracted from the oil fields in the first stage, known as the extraction phase, and is then imported into the refinery along with other inputs deemed stable in the second stage. The desired and undesirable outputs from the second level are separated in this section, and each DMU a distinct two-stage supply chain that is defined in this paper then attempts to increase the desirable output and decrease the undesirable output. The efficiency calculated with the NDEA model is based on the sustainability aspect, which is considered in inputs and outputs during the supply chains.

It is important to highlight that the intended outcomes encompass all usable petroleum derivatives upon completion of the process. This includes final products that are consumable without further processing, such as chemical fertilizers, polyethylene, acids, chemical solvents, and consumable feed. These products are ultimately intended for consumption by end consumers or secondary industrial entities, particularly in sectors such as pharmaceuticals. It is essential to consider the diverse range of output products from petrochemical facilities to ensure comparability, with their value expressed in terms of revenue from product sales in dollar units as the desired outcome. Undesirable output refers to oil waste that is not suitable for consumption and must be reintroduced into the refining process for re-refining or disposed of as waste outside of the cycle. The aim is to minimize these occurrences in subsequent cycles; in addition to considering inputs of a cost nature, the outputs are also evaluated in terms of dollar units within the dataset and supply chain framework.

The dataset of the case study in this paper is taken from Samane Andishmand Pegah (SAP) software. SAP is a knowledge-based company in the fields of consulting, planning, and production of Enterprise Resource Planning (ERP) software. SAP wishes to evaluate and forecast the sustainability of supply chains of the industries. One of the company's areas of operation includes the oil and petrochemical industries. As is seen in *Fig. 5*, each supply chain has two stages: extraction and Refinery. The inputs and outputs of the supply chain are selected based on sustainability concepts and a literature review, which is available in *Table 7* in the Appendix section. These are the factors for assessing efficiency with a sustainability approach. The inputs and outputs of each stage are defined as follows:

Extraction stage: in the extraction stage, we define the inputs and outputs of oil, which are located in 28 locations in Iran.

Inputs

I. Cost of workforce safety (Social responsibility criterion): Costs that are paid to protect and secure workers in the extraction phase (Million $).

II. Personnel salary cost (Economic criterion): Salaries paid for personnel in the extraction stage (Million $).

III. Environmental compensation costs (Environmental criterion): Environmental damage costs (Especially reduction of CO_2 emission) paid by the extraction stage (Million $).

Output

I. Oil: Oil extracted from extraction. (which is the input of stage two and the output of stage one).

II. Refinery stage: In this stage, we define inputs based on sustainability aspects, the same as in stage one.

Inputs

I. Cost of workforce safety (Social responsibility criterion): Costs that are paid to protect and secure workers in the Refinery phase (Million $).

II. Personnel salary cost (Economic criterion): Salaries paid for personnel in the Refinery stage (Million $).

III. Environmental compensation costs (Environmental criterion): Environmental damage costs (Especially reduction of CO_2 emission) paid by the Refinery stage (Million $).

IV. Oil imported to refinery phase (Raw materials; economic criterion): The amount of oil extracted from stage one to the refinery factory.

Output

I. Oil and other salable oil derivatives (Desirable output): Sales volume of desirable outputs (Million $).

II. Oil waste and the negative impact of greenhouse gases (Undesirable output): The volume of undesirable outputs contains waste materials and the impact of CO_2 emission (Million $).

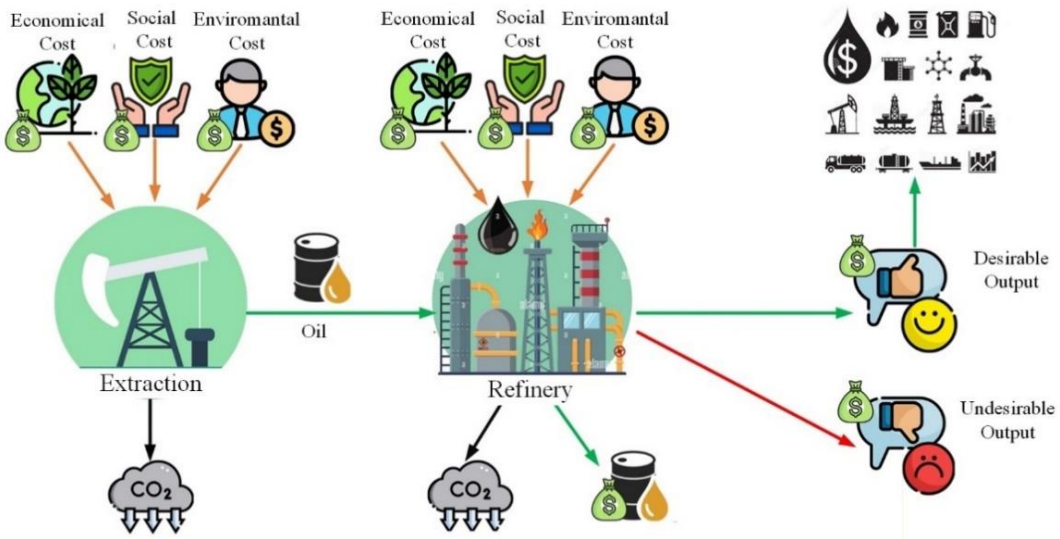


Fig. 5. The structure of supply chains.

The output variables within the network are categorized into two groups: desirable and undesirable outcomes. The desirable output refers to any product that can be utilized for smelting post-exit from the distillation towers or can serve as feed for another industrial unit, representing the total revenue generated from these products within the two-stage supply chain. The undesirable output encompasses all residues and wastes that are either non-utilizable or require re-introduction into the refinery cycle for reprocessing and is also regarded as a total cost incurred. Attention is given to the diversity of products within petrochemical units to enable comparison of the production volume to the consumption input ratio for both desirable and undesirable outputs, measured in dollar units. The crude oil acquired from the extraction process in the first stage is recognized both as the output of the first stage and the input of the second stage, embodying a dual characteristic and serving as an internal variable within the network.

4.2 | Clustering DMUs with Machine Learning

ML plays a crucial role in SCM-related criteria. First, ML can describe the non-linear relationship while traditional methods do not, for the training model of ML better describes how the output (y) changes with input (x). Second, ML is able to deal with unstructured data sets where traditional models fail. For instance, a traditional linear regression model requires a sample size larger than the number of features in the data sets. However, a small company usually fails to gain enough sample size for their SC records. Third, ML and its core constructs are suitable for better SCM performance prediction. If taking the strengths of unsupervised learning, supervised learning, and reinforcement learning into consideration, ML is extremely effective in continually seeking the key factors most affecting SC performance. Fourth, with the help of visual pattern recognition across an SC network, ML is able to explore a lot of potential applications in the maintenance of physical assets and physical inspection. According to *Fig. 2* and based on the literature review, the ML algorithms frequently used in SCM and figure out the SCM activities most suitable for ML. We found a gap that addressed the lack of use of ML algorithms for clustering DMUs in SCM and SSCM in particular. Some of the applicable use cases of ML in SCM have been listed below:

I. Demand forecasting: ML algorithms can analyze historical sales data, market trends, and other relevant factors to predict future demand for products. This helps companies optimize their inventory levels and reduce stockouts or overstock situations.

II. Inventory management: ML can be used to optimize inventory levels by analyzing various factors such as lead times, demand variability, and supplier reliability. This helps in minimizing holding costs while ensuring product availability.

III. Supplier management: ML can analyze supplier performance data to identify patterns and predict potential issues such as late deliveries or quality problems. This can help in improving supplier relationships and reducing supply chain disruptions.

IV. Transportation optimization: ML algorithms can analyze transportation data to optimize route planning, load scheduling, and carrier selection. This can lead to cost savings and improved delivery performance.

V. Warehouse management: ML can be used to optimize warehouse layouts, pick paths, and storage locations, leading to more efficient operations and reduced labor costs.

VI. Risk management: ML can analyze various factors such as geopolitical events, natural disasters, and market trends to predict and mitigate supply chain risks.

In a broader context, ML holds the potential to enhance the decision-making capabilities of SCM practitioners, optimize operational processes, and drive cost-reduction initiatives. The present study aims to delve into the utilization of ML techniques, specifically in the clustering of decision units within the realm of SCM. Python is used for all programming tasks in this section to evaluate and compare the ML algorithms. Following this, three ML algorithms utilized in this study will be introduced, and the results obtained from them will be analyzed.

K-means

K-means is an unsupervised clustering method. This method is able to divide the data into k clusters which would decrease the square errors of each group. K-means is time-and-cost-efficient with its low computational complexity, but the K needs to be preset in advance, which requires prior knowledge of the data sets. What is worse, K-means is prone to being affected by outliers and noise. We should add that K-means works with the concept of maximum similarity between inputs and outputs of DMUs [41].

Hierarchial clustering

Among the unsupervised machine methods is a top-down approach. In this method, each cluster may be placed in more than one cluster because the subset of the larger cluster itself may be at another level of distance. One of the most important points in the implementation of this method is the selection of

appropriate features to start the calculations, which should be given the attention of science researchers in order to avoid increasing the deviation of situations. Finally, according to the characteristics determined by the researcher, the similarity between the data is determined hierarchically, and if the number of categories is determined, the result of clustering will be visible at each level [42].

Deep Embedded Clustering

It is appropriate to use this method when the number of features and dimensions of the data is large and a suitable volume of data is available for the implementation of deep learning. In this method, using a deep neural network, DEC is trained to convert high-dimensional data into a reduced-dimensional form so that it can implement clustering more tangibly. Using this method has a significant improvement in high-dimensional data. Currently, it is used in many sciences that have complex data and need clustering. In the next part, two main parts will be discussed, which are very important in evaluating the results of clustering [43].

4.2.1 | Determining the best number of clusters

In order to ascertain the optimal number of clusters (k value), the Root Mean Square Standard Deviation (RMSSTD) index can be employed. This metric quantifies the level of similarity among members within a cluster. To identify the most suitable value of k, it suffices to compute the RMSSTD for various k values and represent it graphically as a broken line plot. In *Fig. 6*, The graph reveals a decreasing trend in the RMSSTD rate as the value of k increases. Notably, a distinct inflection point resembling an elbow is discernible on the diagram, termed as the Elbow point. Beyond this juncture, marginal reductions in RMSSTD are observed. Consequently, the optimal number of clusters is approximated at the Elbow point, employing this approach to ascertain the most suitable cluster in clustering predicaments. Based on the graph analysis, the number of clusters is inferred to be 4, prompting the implementation of clustering algorithms with k = 4 to validate the research findings.

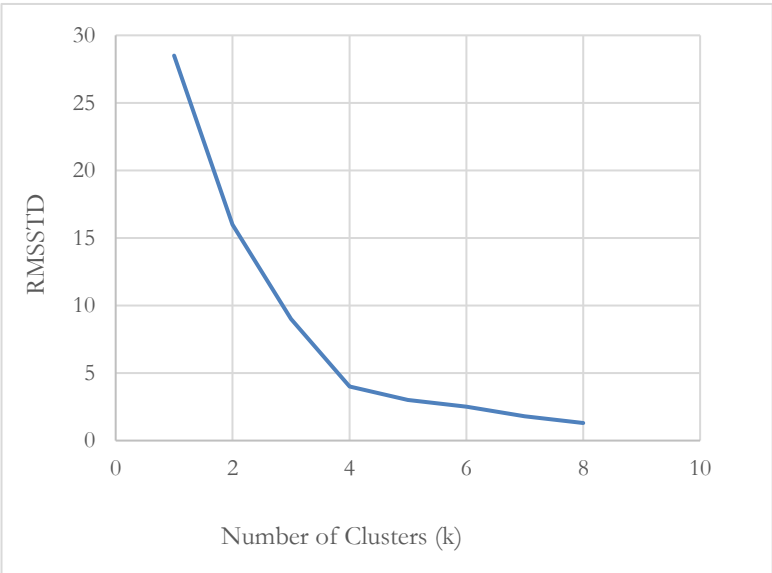


Fig. 6. Elbow point for finding the best number of clusters.

4.2.2 | Evaluation methods of clustering results

Evaluation of the clustering result (Which is an unsupervised learning) is considered an important step in the research process. In general, there are two criteria for evaluating clustering results: 1) internal criteria index, and 2) external criteria index. The internal evaluation index tries to measure the similarity of cluster members and their differences with other clusters. Using the analysis of variance method is one of the evaluation solutions; so based on this statistical criterion, the goal is to obtain a state of clustering that has the minimum

amount of intra-cluster dispersion (That is, the greatest similarity between members) and the maximum amount of inter-cluster dispersion (That is, the greatest difference between other clusters) to be If the number of data dimensions is more than one, it is necessary to calculate the total dispersion between the dimensions under the title of total dispersion. In the following, the three evaluation indicators used in this research will be examined:

I. Silhouette score: the silhouette score is an indicator used for assessing clustering quality in datasets. It determines the similarity of an object to its cluster in comparison to other clusters. The scale of this indicator is [-1, 1]. A high score indicates good matching within the cluster but poor matching with neighboring clusters, while a low score suggests the opposite. A higher silhouette score indicates more distinct clusters, making it a valuable tool for evaluating clustering algorithms and determining the ideal number of clusters for a dataset.

II. Davies-Bouldin score: To calculate this indicator, it is necessary to calculate the two measures of dispersion size and dissimilarity between clusters. According to this criterion, the more dispersion there is inside the clusters, the greater its value, and the smaller the distance between two clusters, the greater its value. In fact, the average value of this criterion will be the maximum ratio of dispersion within to dispersion between clusters, and the lower the value, the higher the quality of the clustering operation.

III. Calinski-Harabasz score: It is the ratio of variance to the ratio between intra-cluster dispersion and inter-cluster dispersion. In other words, it is also called Variance Ratio Criterion (VRC).

At the end of this section, first, the results of clustering by the three introduced algorithms will be displayed. Then, the results obtained based on three evaluation indicators will be examined. In the next step, after determining the best clustering algorithm for the dataset of the case study, the relative efficiency of DMUs placed in each cluster based on the network data envelopment model will be calculated and compared with the non-clustering state. In the realm of data science, a pivotal procedure undertaken by experts involves the meticulous selection of an optimal clustering algorithm through visual scrutiny of the data space. By closely examining the scatter plots and data structures, the range of software implementation choices is refined to attain the most effective solution.

Nevertheless, in contemporary scenarios marked by escalating data dimensions, the visualization of a graph incorporating multiple features often transcends the conventional two or three-dimensional space, rendering the depiction of a dataset infeasible. In such instances, the employment of dimensionality reduction techniques like Principal Component Analysis (PCA) is recommended. It is imperative to note that the application of this solution necessitates tailored programming within the database to align with its distinctive variables and attributes.

The data of 28 two-level supply chains in petrochemical industries has been collected for 90 time periods. The best clustering quality in these 90 time periods belongs to the period in which the quality assessment indices have the best values among all periods. The *Table 2*, the data of the best period between 90 time series, in addition to the results of three ML clustering algorithms, has been illustrated. According to the findings presented in the first three columns of *Table 2*, the outcomes of the three ML subset algorithms have been successfully implemented. The variation in cluster numbering can be attributed to the nature of clustering falling within the domain of unsupervised learning, where there are no predefined labels to delineate units within this domain. In *Table 2*, at each stage, three cost data points with a focus on sustainability indicators are regarded as significant. The revenue generated from light oil serves as an intermediary factor between stages 1 and 2, denoted in millions of dollars, while the income of the industrial unit, acquired as the desirable output at the conclusion of stage 2, is placed alongside the undesirable output. To ensure comparability with the quantity of desirable outputs, predominantly oil and gas products, two additional columns have been included based on tonnage. It is important to note that these tonnage columns are excluded from the calculations of relative efficiency.

Table 2. The data of the best period between all and the results of ML Clustering algorithms.

K-Means	Hierarchial	DEC	DMU	Extraction			Refinery							
				Env Cost	Social Cost	Eco Cost	Light Oil (Million$)	Env Cost	Social Cost	Eco Cost	Oil & Others (Million$)	Undes Output (Million$)	Gas (Tonnage)	Oli (Tonnage)
2	0	2	1	65.1	24.1	16.9	3797.2	87.4	21.4	12.5	4594.7	265.8	22.3	106009.8
1	1	1	2	55.3	34.8	8.9	2856.0	73.1	12.5	14.3	3455.7	199.9	32.1	119769.0
3	3	3	3	81.1	26.7	8.9	3545.0	39.2	25.0	11.6	4289.4	248.1	18.7	129645.4
1	1	1	4	82.0	40.1	13.4	4442.6	52.6	11.6	15.2	5375.5	311.0	16.9	122267.5
2	0	2	5	23.2	36.5	5.3	5079.0	32.1	23.2	14.3	6145.6	355.5	19.6	109307.0
2	0	2	6	29.4	36.5	14.3	3318.6	61.5	34.8	9.8	4015.5	232.3	21.4	99820.1
0	2	0	7	33.9	32.1	7.1	6080.0	35.7	37.4	13.4	7356.8	425.6	25.0	89642.5
3	3	3	8	39.2	19.6	15.2	2390.7	58.8	33.0	6.2	2892.7	167.3	21.4	129864.7
0	2	0	9	80.2	17.8	6.2	2713.3	81.1	12.5	10.7	3283.1	189.9	17.8	85289.9
1	1	1	10	35.7	38.3	4.5	5245.7	67.7	10.7	8.0	6347.3	367.2	17.8	119698.6
2	0	2	11	53.5	17.8	6.2	3512.9	26.7	33.9	16.9	4250.6	245.9	25.8	106004.5
1	1	1	12	82.0	16.0	7.1	3975.5	63.3	12.5	8.9	4810.4	278.3	22.3	123130.4
0	2	0	13	17.8	18.7	13.4	5126.3	53.5	18.7	15.2	6202.8	358.8	27.6	94085.1
2	0	2	14	35.7	39.2	15.2	2482.5	29.4	14.3	5.3	3003.8	173.8	28.5	106332.5
1	1	1	15	58.8	39.2	5.3	4484.5	74.9	40.1	16.9	5426.2	313.9	22.3	123735.6
2	0	2	16	26.7	33.9	8.0	4132.4	49.9	25.8	6.2	5000.2	289.3	21.4	101438.9
3	3	3	17	33.9	16.9	14.3	2625.1	61.5	32.1	4.5	3176.4	183.8	18.7	131593.0
1	1	1	18	85.6	37.4	8.0	2235.6	50.8	10.7	13.4	2705.0	156.5	22.3	119939.3
1	1	1	19	32.1	18.7	14.3	3160.8	66.0	19.6	11.6	3824.6	221.3	19.6	118795.6
3	3	3	20	77.5	25.0	15.2	4460.4	25.0	29.4	13.4	5397.1	312.2	17.8	133622.7
2	0	2	21	35.7	26.7	11.6	2441.5	15.2	20.5	5.3	2954.2	170.9	18.7	108709.8
3	3	3	22	70.4	8.0	8.9	4710.9	74.0	16.0	6.2	5700.2	329.8	20.5	132451.4
1	1	1	23	52.6	40.1	15.2	5550.6	73.1	14.3	12.5	6716.2	388.5	21.4	115863.0
2	0	2	24	17.8	9.8	5.3	2981.6	74.9	22.3	13.4	3607.8	208.7	21.4	99852.2
1	1	1	25	53.5	35.7	15.2	4544.2	61.5	12.5	8.0	5498.5	318.1	22.3	125234.0
1	1	1	26	18.7	16.0	8.9	4526.4	40.1	12.5	10.7	5476.9	316.8	23.2	122051.8
2	0	2	27	81.1	14.3	16.9	2826.5	74.9	40.1	13.4	3420.1	197.9	21.4	103716.3
3	1	3	28	73.1	29.4	7.1	4849.9	64.2	37.4	4.5	5868.4	339.5	25.0	127766.4

The results pertaining to the assessment of clustering quality using the three proposed algorithms are presented in *Table 3*. According to the findings of *Table 3*, the Deep Embedded Clustering (DEC) algorithm demonstrates superior performance across all three metrics, namely the Silhouette score, Davies-Bouldin Score, and Calinski-Harabasz Score. It is important to highlight that within the Python programming segment related to the clustering of DMUs, 28 supply chains within the industrial units under investigation were treated as a black boxes to ensure comparability. Subsequently, the efficiency calculation using the NDEA structure, as depicted in *Fig. 4*, was the focal point of post-clustering analysis.

The inquiry arises as to why clustering based on increased similarities among units will transpire? The research underscores the utilization of clustering as a focal point. This study aims to render inefficient DMUs comprehensible at a more accessible level to enhance and advance their developmental indicators. Through this approach, each unit can strategize its industrial unit development plan based on the best DMU in each cluster and explain its development strategies. Efficient DMUs in each cluster can make their development plan with a target of the most efficient DMU between all efficiency calculations. Finally, all the inefficient DMUs are able to plan a more achievable development plan in relation to the volume of production and the amount of inputs and outputs.

Table 3. Quality assessment between algorithms.

DMU		1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	Score
K-Memans	Silhouette	2	1	3	1	2	2	0	3	0	1	2	1	0	2	1	2	3	1	1	3	2	3	1	2	1	1	2	3	0.607
	Davies-B	2	1	3	1	2	2	0	3	0	1	2	1	0	2	1	2	3	1	1	3	2	3	1	2	1	1	2	3	0.475
	Calinski-H	2	0	3	0	2	2	1	3	1	0	2	0	1	2	0	2	3	0	0	3	2	3	0	2	0	0	2	3	142.1
Hierarchial	Silhouette	0	1	3	1	0	0	2	3	2	1	0	1	2	0	1	0	3	1	1	3	0	3	1	0	1	1	0	1	0.585
	Davies-B	0	1	3	1	0	0	2	3	2	1	0	1	2	0	1	0	3	1	1	3	0	3	1	0	1	1	0	1	0.472
	Calinski-H	0	1	3	1	0	0	2	3	2	1	0	1	2	0	1	0	3	1	1	3	0	3	1	0	1	1	0	1	126
DEC	Silhouette	2	1	3	1	2	2	0	3	0	1	2	1	0	2	1	2	3	1	1	3	2	3	1	2	1	1	2	3	0.644
	Davies-B	1	2	0	2	1	1	3	0	3	2	1	2	3	1	2	1	0	2	2	0	1	0	2	1	2	2	1	0	0.368
	Calinski-H	0	1	3	1	0	0	2	3	2	1	0	1	2	0	1	0	3	1	1	3	0	3	1	0	1	1	0	3	170

4.3 | Assessing Efficiency with NDEA

In this section, efficiency calculations were conducted using the equations outlined in *Figs. 3* and *4*, employing NDEA within the GAMS software. The efficiency results presented in *Fig. 7* encompass all petrochemical industrial DMUs in Iran without factoring in clustering at first. Within *Fig. 7*, under the assumption of CRS , efficiency equations were derived in two scenarios: one without undesirable output and the other incorporating undesirable output. The comparison of total efficiency across these equations was undertaken through two distinct approaches:

I. Multiplicative method: In this case, the total efficiency is equal to the multiplication of the efficiency of stage 1 by stage 2. In this method, in order to ensure the stability of the obtained values, all performance values will be calculated on the condition that the values calculated in the previous stage are constant.

II. Weighted average method: In this method, the total efficiency is equal to the weighted average of the efficiency of stage 1 and stage 2. The weight of each stage is equal to the impact of the inputs. It should be noted that the impact of each stage can be adjusted by managers based on the importance of each stage in each period that determines the outputs.

Table 4. Comparison of clustering on overall efficiency in each cluster.

						Multiplicative				Average			
	The Results After Clustering					Overall Efficiency		Overall Efficiency with Undesirable Outputs		Overall Efficiency		Overall Efficiency with Undesirable Outputs	
	DMU	Extraction	Refinery	Extraction	Refinery	Before Clustering	After Clustering	Before Clustering	After Clustering	Before Clustering	After Clustering	Before Clustering	After Clustering
Cluster 1	1	0.52	1.00	0.52	1.00	0.46	0.52	0.46	0.52	0.73	0.76	0.73	0.66
	5	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1	1	1	0.9
	6	0.56	0.59	0.56	0.60	0.24	0.33	0.26	0.34	0.5	0.575	0.52	0.52
	11	0.88	1.00	0.88	1.00	0.58	0.88	0.61	0.88	0.77	0.94	0.78	0.84
	14	0.36	0.96	0.36	0.97	0.29	0.35	0.29	0.35	0.65	0.66	0.65	0.57
	16	0.76	1.00	0.76	1.00	0.70	0.76	0.70	0.76	0.85	0.88	0.85	0.78
	21	0.39	0.85	0.39	0.88	0.35	0.33	0.36	0.34	0.67	0.62	0.68	0.55
	24	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1	1	1	0.9
	27	0.65	0.96	0.65	0.97	0.15	0.62	0.18	0.63	0.4	0.81	0.42	0.72
Cluster 2	7	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1	1	1	0.9
	9	0.72	0.79	0.72	0.80	0.27	0.57	0.59	0.58	0.53	0.75	0.79	0.68
	13	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1	1	1	0.9
	2	0.45	0.49	0.45	0.86	0.40	0.22	0.40	0.39	0.7	0.47	0.7	0.57
	4	0.53	0.92	0.53	0.99	0.46	0.49	0.46	0.52	0.73	0.73	0.73	0.67
	10	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1	1	1	0.9
	12	1.00	1.00	1.00	1.00	0.65	1.00	0.80	1.00	0.81	1	0.9	0.9
	15	0.80	0.58	0.80	0.60	0.40	0.46	0.41	0.48	0.65	0.69	0.65	0.64
	18	0.35	0.78	0.35	1.00	0.32	0.27	0.32	0.35	0.66	0.57	0.66	0.58
	19	0.60	0.53	0.60	0.95	0.28	0.32	0.36	0.57	0.51	0.57	0.61	0.68
	23	0.62	0.88	0.62	0.88	0.51	0.55	0.51	0.55	0.73	0.75	0.73	0.67
	25	0.53	0.96	0.53	1.00	0.49	0.51	0.49	0.53	0.75	0.75	0.75	0.67
	26	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1	1	1	0.9
Cluster 3	3	0.66	0.79	0.66	0.86	0.55	0.52	0.55	0.57	0.78	0.73	0.78	0.68
	8	0.80	0.50	0.80	0.95	0.19	0.40	0.32	0.76	0.44	0.65	0.6	0.78
	17	1.00	1.00	1.00	1.00	0.49	1.00	0.49	1.00	0.75	1	0.75	0.9
	20	0.83	0.97	0.83	0.98	0.52	0.81	0.52	0.81	0.76	0.9	0.76	0.81
	22	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1	1	1	0.9
	28	1.00	1.00	1.00	1.00	0.83	1.00	0.83	1.00	0.92	1	0.92	0.9

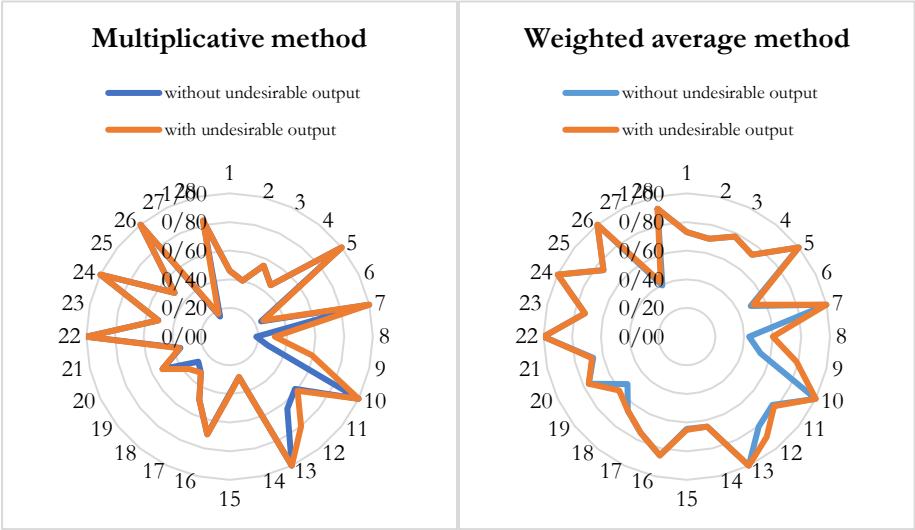
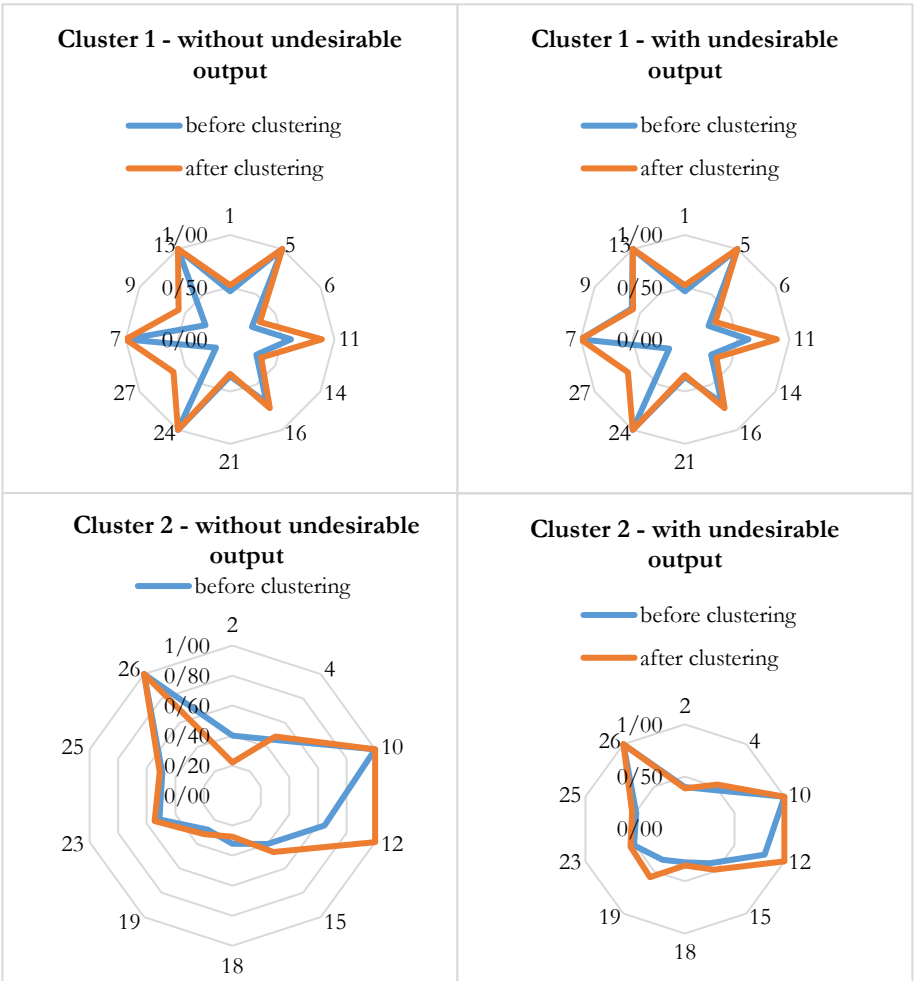


Fig. 7. Graphical comparison of overall efficiency after clustering (in each cluster separately).

The results showed that the calculated total efficiency values increased to a significant percentage after clustering. This could serve as a valuable reference for managers when strategizing production planning for future periods, setting development goals for production metrics, and enhancing productivity. The results presented in *Table 4* can be visually interpreted through graphic diagrams in *Fig. 8*. It was observed that in the multiplicative method, both with and without the presence of undesirable outputs, the overall efficiency values were higher.



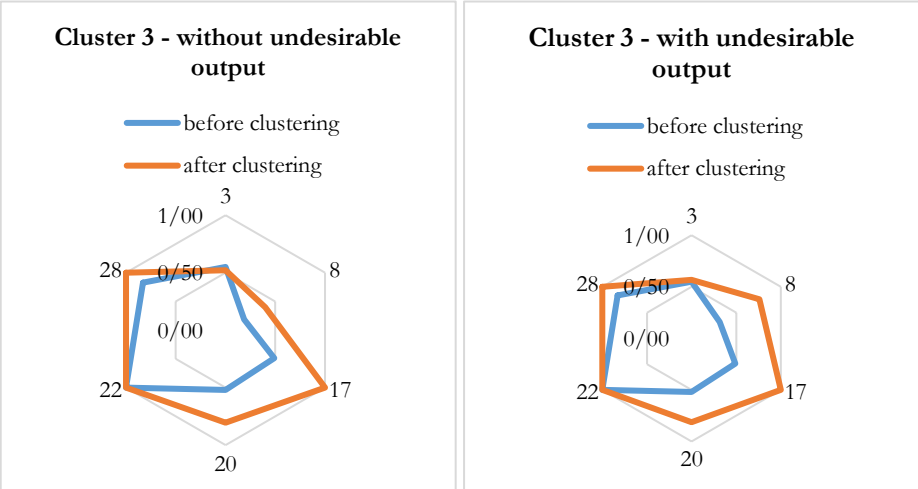


Fig. 8. Graphical comparison of overall efficiency after clustering (in each cluster separately).

5 | Discussion

Engaging in interdisciplinary studies that incorporate innovative problem-solving methodologies grounded in data science and mathematical modeling represents a compelling area of inquiry for researchers. This research examines the application of ML techniques for clustering within the framework of sustainable supply chain principles. Utilizing a two-stage NDEA modeling approach, the study assesses the relative efficiency of the DMUs under consideration. This research focuses on petrochemical units in Iran as a case study. The application of ML for clustering homogeneous units within this sector and the calculation of relative efficiency through a data-driven methodology within each cluster represents a novel contribution to the field in the Iranian context. Detailed information regarding the dataset utilized in this study is provided in the Data Availability section.

The selection of inputs and outputs for each stage of the research is informed by the literature review, specifically detailed in Subsections 2.2 and Section 4. As outlined in the introduction to clustering algorithms, three distinct ML algorithms were implemented utilizing the Python programming language. A significant challenge encountered in this phase was the identification of the optimal clustering algorithm. To address this, three clustering quality evaluation metrics referred to as Silhouette score, Davies-Bouldin score and Calinski-Harabasz score were introduced to assess the clustering quality of the dataset through mathematical calculations. As presented in *Table 3*, an increase in the Silhouette index, alongside a decrease in the Davies-Bouldin and Calinski-Harabasz indices, indicates an improvement in clustering quality. A comprehensive analysis of the nature of each of these indicators is provided in Section 4.2. The findings indicate that the DEC algorithm produces the most advantageous results for the dataset examined in this case study.

In the subsequent section of this research, attention was directed toward modeling utilizing the NDEA two-stage supply chain framework in accordance with sustainability criteria. As highlighted in the literature review, three independent inputs economic costs, social costs, and environmental costs were identified at each stage of the supply chain for conducting relative efficiency calculations within the GAMS. The modeling was conducted under two overarching scenarios: one without the presence of undesirable outputs and another incorporating such outputs, with comparative results illustrated in *Figs. 7, 8* and *Table 4*. This study specifically aims to highlight the impact of enhancing relative efficiency values following the clustering of homogeneous units. Consequently, all metrics were computed in both before and after clustering states to elucidate the effect of clustering on efficiency calculations. To augment this section, a statistical analysis employing hypothesis testing was performed using SPSS, where the means were evaluated through the Paired Sample Test (t-test) method. In this analysis, μ_1 is the mean of efficiencies after clustering, and μ_2 is the mean of efficiencies before clustering. The null and alternative hypotheses are articulated as follows: H_0 (Null

hypothesis) posits that there is no significant difference between the two means, and H_1 (Alternative hypothesis) suggests that a significant difference exists. Based on the SPSS results, the significance level (Sig. 2-tailed) is observed to be less than 0.05, leading to the acceptance of the alternative hypothesis. The equation and detailed outcomes of the hypothesis testing are presented in *Table 5*.

$$H_0 : \mu_1 \leq \mu_2,$$
$$H_1 : \mu_1 \rangle \mu_2.$$

Pair 1: without clustering without undesirable output-with clustering without undesirable output.

Pair 2: without clustering with undesirable output-with clustering with undesirable output.

Table 5. Paired samples test.

	Mean	Std. Deviation	Std. Error Mean	95% Confidence Interval of the Difference		t	df	Sig (2-Tailed)
				Lower	Upper			
Pair 1	-0.0988	0.1639	0.0310	-0.1623	-0.0352	-3.190	27	0.004
Pair 2	-0.1080	0.1539	0.0290	-0.1676	-0.0483	-3.713	27	0.001

Having established the impact of clustering on efficiency calculations, the next step involves identifying the most efficient unit within each cluster. In the proposed methodology, efficiency assessments are conducted for each cluster in accordance with the proposed NDEA two-stage SSC. The most efficient unit in each cluster is regarded as a benchmark for the other inefficient units. This approach provides these units with a more attainable goal compared to efficiency calculations conducted without clustering across all 28 units.

The most efficient units within each cluster will use the top-performing unit among the 28 as a reference point for developing sustainability-based production indicators. Overall, this strategy facilitates a more realistic and achievable development plan for the industrial units. Furthermore, this methodology can be applied across various supply chains to implement a phased development policy, allowing for incremental progress toward enhanced efficiency and sustainability.

The current review of research within the field of data science emphasizes the comparison of execution times among various algorithms as a critical component. However, it is important to note that this study primarily focuses on the quality of clustering and the identification of optimal separation between units. Given the nature of the dataset employed, the computational time required for clustering is relatively minimal and does not serve as a determining factor for identifying the most effective clustering method. Consequently, the aspect of banding is not addressed in this analysis. Nevertheless, the comparison of execution time and the complexity of the Python code implemented in this research is presented in *Table 6*.

Table 6. The time and space complexity of algorithm comparison.

Algorithm	Time Complexity	Space Complexity
K-Means	O(n * k * d * i)	O(n*d + k*d + n)
DEC	O(n) or O(n^2)	O(n)
Hierarchical	O(n^2 log(n))	O(n^2)

For example, the time complexity of the my k-means clustering function is primarily determined by the K-Means clustering algorithm, which has a time complexity of O(n * k * d * i), where n is the number of data points, k is the number of clusters, d is the number of dimensions, and i is the number of iterations. The storage of the input data, the K-Means model, the cluster labels, and the output results determines the space complexity of the function. The input data is stored as a numpy array, which requires O(n * d) space. The K-Means model also requires space for storing the cluster centers and labels, which is typically O(k * d). The cluster labels and output results are stored in lists, which require additional space. Overall, the space complexity can be considered as O(n*d + k*d + n). Analysis of the DEC showed that overall, the time complexity of the DEC_clustering function is dominated by the training of the autoencoder model and the clustering of the encoded data, which are both O(n) or O(n^2) depending on the specific parameters of the

model and the clustering algorithm. The space complexity of the DEC_clustering function is also important to consider. It involves storing the input data, the autoencoder model parameters, the encoded data, and the cluster labels. The space complexity depends on the size of these data structures, but it is usually O(n), where n is the number of samples in the input data. The results showed that DEC still gives better results.

6 | Conclusion and Managerial Implications

The utilization of data science in enhancing and interpreting datasets stands as a crucial element in gaining a competitive edge within the industry, with these datasets being recognized as implicit assets within organizational contexts. The current research study placed particular emphasis on the role of data science in the domain of SCM as part of the research methodology, specifically delving into the utilization of ML algorithms in the domain of clustering, a form of unsupervised learning. In examining contemporary algorithms featured in international articles and publications, this paper explored the significance of such reviews in prior research endeavors concerning the assessment of relative efficiency and sustainability within the supply chain. This investigation aimed to chart a research path in the realm of enhancing supply chain efficiency through the application of NDEA within a data science framework.

The study demonstrated that the optimal dividing of DMUs into clusters based on maximum similarity can offer a more effective practical solution compared to conventional DEA methods for assessing relative efficiency. By clustering DMUs, inefficient units within each cluster were brought closer to improve efficiency by aligning them with the most efficient unit in their respective cluster. This approach may serve as a viable strategy for short-term goal-setting within industrial units to strive towards achieving the efficiency levels of the leading unit in their cluster. Conversely, while relative efficiency has been applied to all DMUs irrespective of clustering, the efficient units within each cluster can aim to achieve the performance metrics of the most efficient unit among the 28 industrial units. This approach allows for the utilization of the methodology proposed in this article to conduct calculations and chart a path for operational enhancement across all units. The findings indicated that through clustering, the inefficiency level of each decision unit decreased, highlighting the necessity for units within each cluster to emulate the performance of the most efficient unit in their respective cluster to attain relative efficiency. It is important to note that by employing DEA in a multiplicative form and giving precedence to stage 2, a significant portion of the weights in the equation were assigned to optimize the efficiency of stage 2. Clustering and introducing a hierarchical improvement strategy can motivate industrial units to engage in continuous enhancement and advancement, a notion supported by the simulation results. In a separate study comparing efficiency calculations with and without undesirable outputs, the findings revealed that considering undesirable outputs as desirable outputs through a linear transformation resulted in increased total efficiency and stage 2 efficiency across all scenarios compared to calculations without undesirable outputs. This method underscores the beneficial impact of incorporating this factor in enhancing the precision of relative efficiency evaluations. Such an approach fosters ongoing improvement and incremental progress within industries, reflecting a key aspect of sustainable development in management principles.

For future research and the continuation of the present study, in addition to exploring modeling techniques such as Dynamic Network Data Envelopment Analysis (DNDEA) with a sustainability approach, further investigation into various clustering methods, including clustering using DEA, could be beneficial using the existing dataset. The DEA-based clustering method involves utilizing piecewise production functions derived from the DEA approach to cluster data based on input and output characteristics. This approach not only assigns DMUs to specific clusters but also considers the type of production function characteristic of each cluster. This method can be instrumental for managerial decision-making, particularly for identifying necessary adjustments in input-output resource combinations to position units within desirable clusters. On the other hand, The design of a Closed-Loop Supply Chain (CLSC) amid uncertainty and partial disruptions poses significant challenges. CLSC aims to integrate reverse logistics into traditional supply chains, promoting sustainability and environmental friendliness. However, such uncertainties can hinder the efficiency and reliability of CLSCs. To tackle these issues, a chance constraint is proposed, enabling the incorporation of

probabilistic uncertainties into decision-making processes. The objective is to create a robust CLSC model that adeptly manages uncertain factors like demand, return rates, and product quality. The study employs the Markowitz approach to address uncertainty in the objective function by blending the mean with the standard deviation coefficient. Findings indicate that integrating uncertainty into the model enhances profitability compared to a deterministic approach, as the uncertain model demonstrates greater responsiveness to fluctuating demands and effectively manages confidence inventory dynamics. While strategic decisions regarding the number of production, distribution, and destruction facilities remain largely unchanged, the capacity of destruction centers in the uncertain model is slightly reduced due to varying product quality. Overall, the inclusion of uncertainty improves the model's transparency and supports better decision-making, leading to increased profitability through enhanced demand responsiveness and inventory management [44].

Furthermore, examining the impact of past efficiency levels on future efficiency could involve utilizing the DNDEA model to analyze two-stage supply chains. This model can account for feedback loops involving the return of waste to the refining cycle and incorporate additional factors into the calculation of relative efficiency. This expanded analysis could provide valuable insights into optimizing supply chain operations and enhance overall efficiency in a dynamic environment.

Another idea is about the efficiency of Industry 4.0 implementations, which utilize advanced computerized systems by employing both classical and fuzzy DEA models. Recognizing the uncertainty in the required data, the authors introduce fuzzy DEA to provide a more accurate evaluation of productivity, allowing for different units of measurement and independence from analytical production functions. This study is notable as the first quantitative analysis of industry 4.0's impact on productivity, demonstrating how fuzzy factors can influence decision-making. A practical application in a home appliances manufacturing company further illustrates the advantages of fuzzy DEA over traditional methods in assessing the effects of Industry 4.0 on machine productivity [45].

Author Contributaion

The first author (Sina Sayardoost): this author had the primary role in idea generation, research design, data collection, formal analysis, and article writing. He has also participated in providing supplementary cases as one of the main authors. The second author (Keikhosro Yakideh): responsible and corresponding author, involved in formal analysis, data verification and review, project planning, and reviewing. The third author (Mahmoud Moradi) contributed to formal analysis for the research and review. The fourth author (Mostafa Ebrahimpour) was involved in writing, reviewing, and editing the article. All authors have read and approved the manuscript. Only individuals who contributed significantly to the reported work are credited as authors.

Data Availability

The dataset of the case study in this paper is taken from Samane Andishmand Pegah software. Samane Andishmand Pegah (SAP) is a knowledge-based company in the fields of consulting, planning and production of ERP software. The dataset which is used in this paper was about the supply chain of petrochemical industries and is available at the following link: (https://b2n.ir/sinasayardoost_Dataset).

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Appendix

Table A1. The factors for assessing sustainability.

Variables	Authors									Category		
	Kiani et al. [22]	Kalantary and Farzipoor [23]	Boudaghi and Farzipoor [5]	Badiezadeh et al. [46]	Yousefi et al. [47]	Izadikhah and Farzipoor [48]	Galal et al. [49]	Ji et al. [50]	Tajbakhsh et al. [51]	Economic variables	Environmental Variables	Social Variables
Transportation cost	*								*	*		
Environmental cost				*							*	
Cargo tonnage/ Passenger numbers											*	
Energy cost			*							*		
Investment			*	*	*			*	*	*		
Operational cost					*			*		*		
Material cost		*	*					*		*		
Staff cost				*	*			*		*		
Fuel consumption	*			*				*		*	*	
CO_2 emission	*			*				*			*	
Operational and Nonoperational employees		*	*	*	*			*	*			*
Quality of product			*		*					*		
Number of green products											*	
People satisfaction												*